

Metropolitan University, Sylhet
Department of Computer Science and Engineering
Term Final Examination – (Spring) 2026
Program: B.Sc. in CSE; Batch: 63 (A, B, C, and D)
Course: GED-119: Engineering Ethics and Cyber Law

Time: 2 Hours

Marks: 40

Answer any **four** questions. All parts of each question must be answered sequentially.

Figures in the right margin indicate full marks.

1. a) Define the term 'ethics' from the perspectives of Aristotle and Immanuel Kant, including its types and basic elements. 4
b) Explain the difference between ethics and engineering ethics and the three key obligations of an engineer under professional responsibility. 4
c) Imagine you are an engineer working on a project involving artificial intelligence. Identify one ethical concern you might face and explain how you would address one ethical concern it using the ethical decision-making process. 2
2. a) Define 'intellectual property' and 'plagiarism', including their types relevant to computer engineering. 4
b) Describe the relationship between intellectual property, plagiarism, and academic integrity. 3
c) Evaluate the consequences of ethical violations related to plagiarism and intellectual property infringement in professional engineering practice. 3
3. a) State when the Cyber Tribunal was established and expanded in Bangladesh, and the key laws under which they are established. 2
b) Explain the jurisdiction and structural composition of the Cyber Tribunal and Cyber Appellate Tribunal in ensuring fairness and accountability in cyber justice. 4
c) A person's social media account is hacked, and false, defamatory content is published. Now, summarize how the Cyber Tribunal handles this case, along with the powers and procedural role of the appellate tribunal. 4
4. a) Define 'Cyber Crime' and 'Cyber Offence' with relevant examples, including the importance of cyber law in protecting users and organizations. 4
b) Explain the basic classification of cybercrimes based on their targets (individual, property, organization, society/state). 3
c) A person spreads fake news online, affecting public order. Now, analyze the type of cybercrime and its impact on society/state. 3
5. Suppose you are **Sarif**, a CSE engineer who develops a digital healthcare system but deploys it without proper security testing, despite being aware of vulnerabilities. He does not disclose these issues and works beyond his area of competence. As a result, the system is hacked, patient data is leaked, and public safety is at risk. Now,
 - i. Identify two relevant ethical principles from the Code of Ethics for Engineers that are violated in this scenario. 3
 - ii. Analyse how Sarif's actions violate the Rules of Practice and Professional Obligations of engineers, including appropriate solutions. 7

6. a) Define 'digital users' and 'digital professionals'. 2
- b) Explain the rights and obligations of digital users, including privacy, data protection, and access to information. 4
- c) A company collects user data without ensuring proper security. Now, identify the violated right and explain the accountability. 4

Metropolitan University
 Department of Computer Science and Engineering
 Spring Term Final Examination – 2026
 Program: B.Sc. in CSE; Batch: 63
 Course: CSE 131: Basic Electronics Engineering

Time: 2 Hours

Marks: 40

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. a) Differentiate p-type and n-type semiconductor materials based on their dopant atoms, majority and minority charge carriers, and conductivity behavior. 4
 b) Determine the I for each of the networks shown below. 6

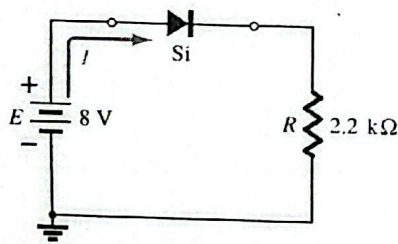


Fig.1b(i)

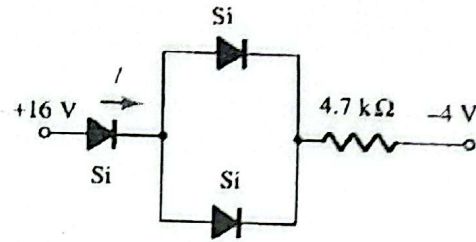
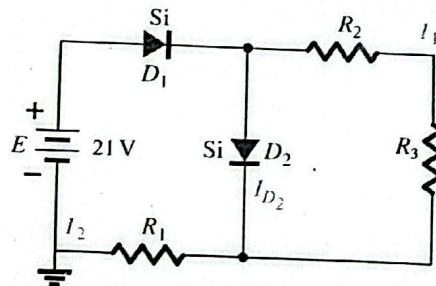


Fig.1b(ii)

2. a) A p-n junction diode is operating at a temperature of 25°C and the applied bias voltage is 0.8 V, the reverse saturation current at this stage is 60 nA, calculate following: 4
 I. Thermal voltage of the diode.
 II. Diode current through the diode if the ideality factor is 2.
 b) Determine the I_1 , I_2 , I_{D2} , V_{R1} , V_{R2} , and V_{R3} for the given network in Fig.2b. 6



Here,
 $R_1 = 4.8 \text{ k}\Omega$
 $R_2 = 0.8 \text{ k}\Omega$
 $R_3 = 0.3 \text{ k}\Omega$

Fig.2b

3. a) Describe the working principle of a bridge rectifier circuit while converting alternating current to direct current. Use appropriate diagrams and waveforms. 4
 b) Determine the output voltage v_o and sketch the output waveform for the network below. If the two ideal diodes are replaced by two germanium diodes, determine the resulting output voltage and sketch the corresponding waveforms. 6

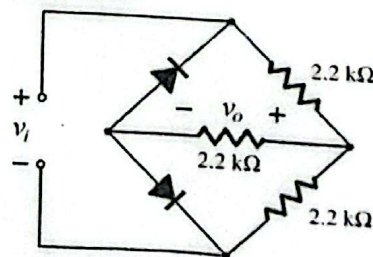
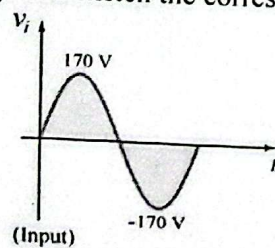


Fig.3b

4. a) Compare series and parallel clipper circuits in terms of their configuration and output characteristics, and explain how diode orientation determines the output waveform. 4
 b) Consider the given signal as AC input to each of the networks shown below. Determine and sketch the output voltage V_{out} for each network. 6

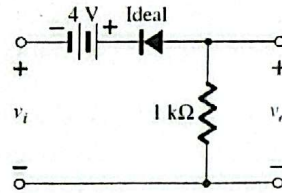
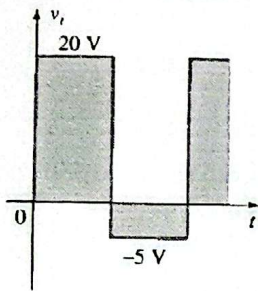


Fig.4b(i)

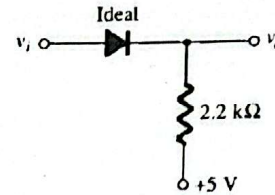


Fig.4b(ii)

5. a) Briefly explain the operation of an n-channel depletion-type MOSFET under different gate voltages, providing proper diagrams and illustrations. 4
 b) I. Illustrate a detailed structure of an enhancement-type P-channel MOSFET with proper terminal notations. 6
 II. Determine V_{GS} and V_{DS} for the circuit shown in Fig.5b. Assume this particular MOSFET has minimum values of $I_{D(on)} = 250$ mA at $V_{GS} = 6$ V and $V_{GS(th)} = 2.5$ V.

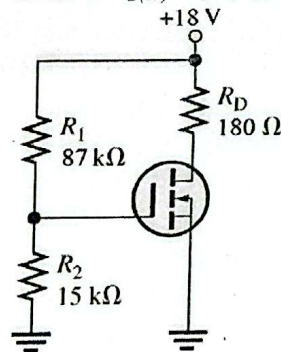


Fig.5b

6. a) Briefly explain the working principle of a npn transistor using necessary diagrams. 4
 b) For the given network in Fig.6b, determine the value of I_B , I_C , I_E , V_{CB} , V_{BE} , and V_{CE} . 6
 Assume the BJT is Si based and β is 140.

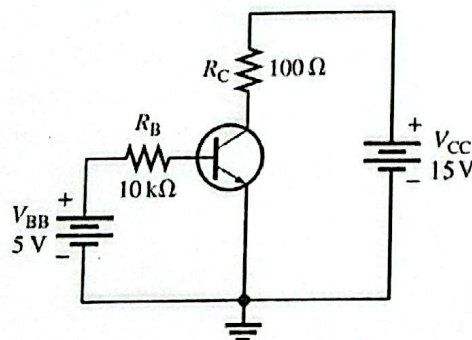


Fig.6b

Metropolitan University
Department of Computer Science and Engineering
Spring Term Final Examination – 2026
Program: B.Sc. in CSE; Batch: 63
Course: CSE 133: Data Structure

Time: 2 Hours

Marks: 40

Answer any four questions. All parts of each question must be answered sequentially.
Figures in the right margin indicate full marks.

1. Consider the array: C, O, M, M, U, N, I, C, A, T, I, O, N
- a) Sort the array in descending order using the **Insertion Sort** algorithm and show each step. Take alphabetical order in account. 5
 - b) Reverse the array from 1(a). Then, Apply **Binary Search** to search 'G' and show each step. 3
 - c) A system receives online orders throughout the day, but the total number of orders is unknown in advance. The system must store each order as it arrives. Which memory management approach should be used? How does this approach handle continuous data growth? 2
2. a) Convert the given **Infix** expression into a **Postfix** expression using a **Stack** and show each step: 5
- $(A+B^D) / E-F * (G+H/K) + C$
- b) Evaluate the following **Prefix** Expression using a **Stack** where, $A = 12, B = 7, C = 3, D = 2, E = 1, F = 5$ 2
- $+ / A - BC * D + EF$
- c) Consider the following queue of characters, where **QUEUE** is a circular array (0-based indexing) which is allocated four memory cells:
 $FRONT = 2, REAR = 0$ **QUEUE:** F, , D, E 3
Perform the following operations and show front and rear value at each step (Write overflow/underflow if they occur in any case).
- | | | |
|----------------|----------------|---------------|
| 1. Enqueue (L) | 2. Enqueue (P) | 3. Dequeue () |
| 4. Dequeue () | 5. Enqueue (T) | 6. Dequeue () |
3. a) A **BST** stores city names in alphabetical order. Insert: 5
- "Dhaka", "Chittagong", "Rajshahi", "Barisal", "Sylhet", "Khulna"**
- (i) Construct the **BST**.
 - (ii) Search for **"Comilla"** and explain why the search fails.
 - (iii) Delete **"Dhaka"** and draw the updated **BST**.

b) An array represents a binary tree in level-order as follows:

A, B, C, D, E, F, G

- (i) Construct the binary tree.
- (ii) Write the Inorder traversal.
- (iii) Write the Preorder traversal.

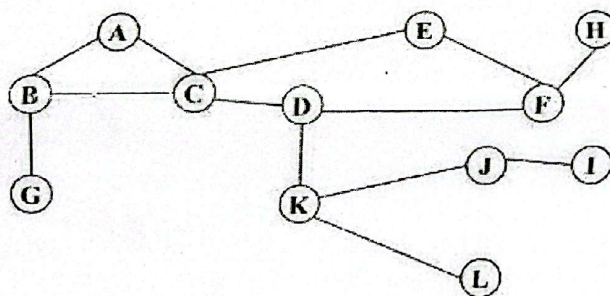
4. a) A singly linked list is initially **empty**. The following operations are performed in sequence: 7

- I. Insert 10 at beginning
- II. Insert 20 at end
- III. Insert 15 after 10
- IV. Delete node containing 20
- V. Insert 25 at end
- VI. Insert 5 at beginning
- VII. Delete node containing 15

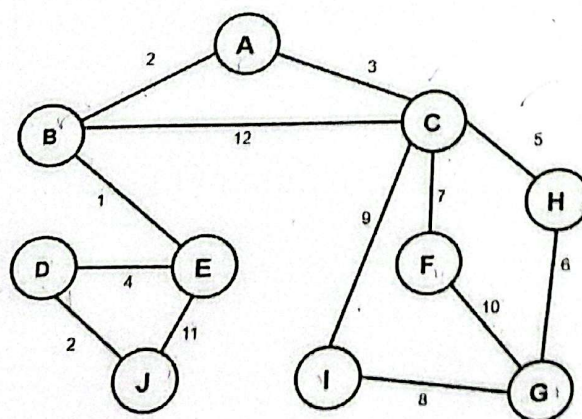
Assume that nodes are allocated at the following memory addresses in the order of creation: 1000, 2000, 3000, 4000, 5000. Show the linked list after each operation.

b) State two advantages and two disadvantages of linked list. 3

5. Traverse the following graph in BFS and DFS order. Show the implementation process using stack or queue. Assume that A is the starting node. 5+5



6. a) Apply **Kruskal's algorithm** to find the Minimum Spanning Tree (MST) for the graph. Calculate the total weight of the MST. 5



b) Find the shortest path using the **Dijkstra algorithm** where the source vertex is A using the graph from 6(a). 5

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Program: B.Sc. in CSE; Batch: 63

Course: MAT 123: Differential Equations and Laplace Transforms

Time: 2Hours

Marks: 40

Answer any **four** questions. All parts of each questions must be answered sequentially.
Figures in the right margin indicate full marks.

1. (a) Define the Laplace transform of a function $F(t)$. If $L\{F(t)\} = f(s)$, prove that, 3
 $L\{F''(t)\} = s^2 f(s) - sF(0) - F'(0)$
- (b) Evaluate $L^{-1}\left\{\frac{3s-6}{s^5} - \frac{s-15}{s^2-25} + \frac{4}{(s-2)^2+16}\right\}$ 3
- (c) Evaluate the following inverse Laplace transform, 4

$$L^{-1}\left\{\frac{5s+3}{(s-1)(s^2+2s+5)}\right\}$$
2. (a) Find the Laplace transform of $e^{4t} + 4t^3 - 2\sin 3t + 3\cos 5t + e^{3t}t^4$. 4
- (b) Solve the following differential equation by using Laplace transform: 6
 $y'' - 3y' + 2y = e^{2t}; \quad y(0) = -3, y'(0) = 5$
3. (a) Define differential equations with example. 1
- (b) Determine the order, degree and check if the equations are linear or nonlinear. 4
 - (i) $\left(\frac{d^2y}{dx^2}\right)^3 + 2\frac{dy}{dx} + y\sin x = 0$
 - (ii) $\left(\frac{d^2y}{dx^2}\right)^2 = \left(\frac{dy}{dx}\right)^4 + 1$
 - (iii) $\sqrt[3]{\left(\frac{d^2y}{dx^2}\right)^4} + y = \sqrt[5]{\left(\frac{dy}{dx}\right)} + y^2 + x^2$
 - (iv) $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^4 z}{\partial y^4} = z$
- (c) Test whether the curve $y = (x^2 + c)e^{-x}$ is a solution or not of the differential equation $\frac{dy}{dx} + y = 2xe^{-x}$. Form a differential equation of the family of curves $y = e^x (A\sin x + B\cos x)$; by eliminating the constants A and B. 5
4. (a) Consider the following DE: $(y^2 + 2xy)dx - x^2dy = 0$ 6
 - (i) Test whether the above equation is exact or not.
 - (ii) Find an integrating factor of the form y^n , where n is a positive integer
 - (iii) Multiply the given equation by the integrating factor found in (ii) and solve the resulting exact equation.
- (b) Find the general solution of the following differential equation 4

$$\frac{dy}{dx} + \frac{2x+1}{x(x+1)}y = \frac{x-1}{x}; \quad y(1) = 2$$

5. Determine the solution of the given differential equations

(a) $(x + 4)(y^2 + 1)dx + y(x^2 + 3x + 2)dy = 0;$ $y(0) = 1$ 5

(b) $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 25y = 0;$ $y(0) = -3, y'(0) = -1$ 5

6. (a) Solve the differential equation using the homogeneous equation approximation: 6

$$(2xy + y^2)dx - (2xy + x^2)dy = 0; \quad y(2) = 1$$

(b) Define the Bernoulli differential equation. Solve the following differential equation: 4

$$dy + (4y - 8y^{-3})xdx = 0;$$